Auteur: Stan Van Campenhout
Studierichting: Informatica
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$$\int x^n \ln(x)\,dx = \frac{1}{n+1}\int \ln(x)\,d\left(x^{n+1}\right) = \frac{x^{n+1}}{n+1}\ln(x) - \frac{1}{n+1}\int x^{n+1} d\left(\ln(x)\right)$$
$$= \frac{x^{n+1}}{n+1}\ln(x) - \frac{1}{n+1}\int x^n dx$$
$$= \frac{x^{n+1}}{n+1}\ln(x) - \frac{x^{n+1}}{(n+1)^2} + c$$

References